EE605 Quiz 1 Solutions

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1 Question 1

Let F be an arbitrary set (the restriction that F be a finite field is not necessary to show that the Hamming distance $d_H(\cdot,\cdot)$ is a metric on F^n).

1.a

Let $x, y \in F^n$. Since $d_H(x,y)$ is the number of positions where x and y differ, clearly, $d_H(x,y) \geq 0$ (it does not make sense for x and y to differ in *less than zero* positions!). Now, let $d_H(x,y) = 0$. Then x and y must differ in no positions i.e. they must be elementwise equal. So they are equal. Conversely, if they are equal, they must be elementwise equal, so they cannot differ in any position i.e. the Hamming distance between them must be zero.

1.b

Trivially, it is symmetric. The number of positions x and y differ is equal to the number of positions y and x differ.

1.c

Let $x, y, z \in F^n$. We want to show that $d_H(x,y) \leq d_H(x,z) + d_H(z,y)$. We may write

$$d_H(x,y) = \sum_{i=1}^{n} \mathbb{1}_{x_i \neq y_i}$$

where $\mathbb{1}_{x_i \neq y_i} = 1$ if $x_i \neq y_i$ and 0 otherwise, and instead prove the inequality for these summands on each side. In case $x_i = y_i$, we are done, since the corresponding summand on the other side cannot be negative. In case $x_i \neq y_i$ (i.e. the summand is 1), it cannot be the case that both $x_i = z_i$ and $z_i = y_i$. So at least one of these equalities must not hold, in which case the corresponding summand is atleast one. We are done.

2 Question 2

There are many ways to do this. Multiplying equation (2) with 2 and subtracting from it equation (1) gives $(2\cdot 2 - 1)y = (2.3 - 1)$. In $GF(5)$, $2 \cdot 2 = 4$ and $2 \cdot 3 = 1$. So we get $4y = 0$. Multiplying by 4 gives $4 \cdot 4y = 4 \cdot 0 \Rightarrow y = 0$. Substituting this in equation (2) gives $x = 3 - 2 \cdot 0 = 3$. So the solution is $x = 3, y = 0$.

3 Question 3

3.a

$$\begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 1 & 1 & 0 & 3 & 1 \\ 0 & 1 & 1 & 1 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 3 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 3 & 2 & 1 \\ 0 & 0 & 3 & 4 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow 2 \cdot R_3} \begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 3 & 2 & 1 \\ 0 & 0 & 1 & 3 & 2 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 - 3 \cdot R_3} \begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 & 2 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2 \cdot R_3} \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 & 2 \end{pmatrix}$$

which is in the form $(I \mid A)$ i.e. systematic form.

3.b

For a generator matrix G in the form $(I \mid A)$ of a linear code C, a parity check matrix for C is $(-A^T \mid I)$. Here,

$$A = \begin{pmatrix} 0 & 1 \\ 3 & 0 \\ 3 & 2 \end{pmatrix}$$

So a parity check matrix is

$$\begin{pmatrix} -0 & -3 & -3 & 1 & 0 \\ -1 & -0 & -2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 & 2 & 1 & 0 \\ 4 & 0 & 3 & 0 & 1 \end{pmatrix}$$

3.c

For a linear code C, its minimum distance is d if and only if some d columns of a parity check matrix H of C are linearly dependent and every set of $d-1$ columns of H is linearly independent. Observe that the 2^{nd} and 4^{th} columns of H are scalar multiples of each other; they are not linearly independent. Also, every set consisting of one column of H is clearly linearly independent since H does not have an all-zero column. So $d(C) = 2$. Now, G is a parity check matrix of $C^{\perp}$. Let c_i denote the i^{th} column of G. From the systematic form of G, we see that $c_1 + 2c_3 = c_5$. Further, it is easily seen that no two columns of G are scalar multiples of each other. So $d(C^{\perp}) = 3$.

4 Question 4

4.a

C is a (6, 3) linear binary code. The systematic generator matrix G is obtained as follows.

u_1	u_2	u_3	$v_1(= u_1 + u_2)$	$v_2(= u_2 + u_3)$	$v_3(= u_1 + u_3)$
1	0	0	1	0	1
0	1	0	1	1	0
0	0	1	0	1	1

Therefore, $G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [I_3|A]$

A parity check matrix of the code can be obtained as

$$H = [-A^T|I_3] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

4.b

The set of codewords {000000, 100101, 010110, 001011, 110011, 011101, 101110, 111000} of C is obtained from the generator matrix G. Observe that there are 4 codewords of weight 3, and 3 codewords of weight 4.

Note that an undetected error occurs when the error pattern is a codeword. Therefore, given that the code is used over a BSC($p = \frac{1}{4}$) channel, the probability of undetected error is given by

$$\begin{aligned} P_{ue}^C &= 4p^3(1-p)^{6-3} + 3p^4(1-p)^{6-4} \\ &= 4p^3(1-p)^3 + 3p^4(1-p)^2 \\ &= \frac{4*27+3*9}{4096} = \frac{135}{4096} = 0.033 \quad \text{for} \quad p = \frac{1}{4} \end{aligned}$$

4.c

The corresponding syndrome table is given below.

Coset leader (e)	Syndrome ($s = He^T$)
000000	000
000001	001
000010	010
000100	100
001000	011
010000	110
100000	101

4.d

The received vector is $r = (011001)$. The syndrome is obtained as $s = Hr^T = 100$, with corresponding error pattern e = 000100. Therefore, the received vector is decoded as $c = r - e = r + e = (011101)$.